
Avionics Wire Prep Master Class

Presented by Bill Randall

Agenda

- Risk & Importance of Wire Stripping
- Identification
- Inspection
- Tooling & Application



Wire Stripping

Risk & Importance of Wire Stripping

After painstakingly reassembling the wreckage of TWA Flight 800 in 1996, the NTSB concluded the center-wing fuel tank had ignited, most likely after a short circuit in a wire bundle which led to a spark in the fuel-gauge sensor.



From this study a new acronym was coined which is widely used in the Aerospace Industry “EWIS” Electrical Wiring Interconnection Systems (EWIS).

Wire Stripping

Risk & Importance of Wire Stripping

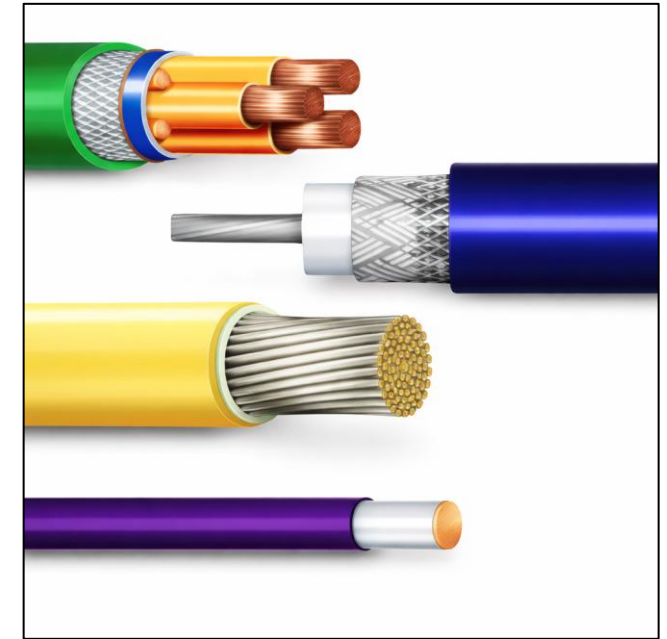
When working with aircraft wiring, proper techniques are crucial for even the simplest tasks, such as stripping insulation from a wire. This ensures the quality and functionality of the installation. The most important aspects of stripping wire are:

- Identify the wire type and choose the proper stripping tool.
- If applicable, adjust the tool to avoid nicking, cutting, or otherwise damaging the strands.
- Do not strip off too much insulation.
- Inspect the conductor for any damage after stripping.

Wire Stripping

Identification

- Aerospace wiring involves high-performance cables focusing on lightweight, durable, and reliable connections.
- Aerospace manufacturers use specialized wires for electrical conductivity, incorporating insulated materials to combat areas of high temp, extreme environments, and anti-chaffing.
- Military specification and industry standards have been created for critical applications in avionics, flight controls and data transmission.
- SAE-AS5768 covers the design and performance requirements to include tools and associated accessories used to strip wires in the aerospace and aviation industry.



Wire Stripping

Identification

Wire and cable insulation materials vary widely, each offering properties tailored to specific applications. In the aerospace industry, some of the most commonly used materials include:

- **Tefzel** (BMS 13-60, MIL-W-22759): Preferred, lightweight, excellent thermal/chemical resistance, good for hot/cold conditions, often the standard.
- **Polyimide (PI)** (MIL-W-81044): High-temperature rated, used in demanding applications, often with other materials.
- **Crosslinked Polyalkene (XLETFE)** (BMS 13-48): Another common lightweight, durable type.
- **MIL-W-5086/7078** Older standard, nylon-coated, resistant to hydrocarbons (fuel/oil).
- **MIL-W-16878** General-purpose, medium, and high-temperature hookup wires, with PTFE or FEP insulation.



Wire Stripping

Identification

- American Wire Gauge (**AWG**) is a standardized system for sizing electrical wires in North America.
- A higher AWG number indicates a smaller wire diameter with the smallest common size being 32 AWG.
- A lower number (like 0, 00, 000, 0000) signifies a larger, thicker wire, and largest 0000 AWG (4/0).
- This system defines wire dimensions, with specific gauges carrying different current capacities, crucial for preventing overheating and ensuring safety, as thicker wires (lower AWG) handle more current.



AWG WIRE SIZE CHART



Wire Stripping

Identification

Mil-Aero wire **Base Part numbers** are primarily designated by **MIL-W-22759** typically used in aerospace, defense, and high-temperature applications.

M22759/ 11 - 22 - 9

Base Part number

Slash sheet

Gauge

Color

- Mil-Aero wire **slash sheets** define specific technical requirements (insulation, voltage, temperature) for aerospace and military-grade electrical cables.
- Specifications for M22759 **gauge range** from the largest of 4/0 through the smallest of 30 gauge.
- MIL-STD-681 defines **color coding** for military specification wires using a 0-9 numbering system for base colors and stripes.

0-Black, 1-Brown, 2-Red, 3-Orange, 4-Yellow, 5-Green, 6-Blue, 7-Violet, 8-Gray, 9-White

Wire Stripping

Identification: Mil Aero Wire Application Guide:

DMC recommends the use of “AS” qualified wire strippers and blades for aerospace and critical electrical wiring interconnection systems (EWIS). AS5768 defines wire stripping tools used by major OEM’s, aircraft and avionics manufacturers.

TOOL PART NUMBER AS5768/2-T	BLADE PART NUMBER AS5768/2-B	WIRE TYPE	WIRE SIZE CAVITIES
1	1	AS22759/9, /10, /11, /12, /13-/15 MIL-DTL-81381/3, /4	16-22
2	2	AS22759/9, /10, /11, /12, /13-/15 MIL-DTL-81381/3, /4	24-30
3	3	AS22759/32, /34, /35, /41-/43 MIL-DTL-81381/1, /2, /6-/14, /17-/22	24-30
4	4	AS22759/32, /34, /35, /41-/43, /59, /60	16-24
5	5	AS22759/80, /81, /91, /92, /180-/182, /191, /192	16-24
6	6	AS22759/80, /81, /91, /92, /180-/182, /191, /192	20-26
7	7	AS22759/83-/90, /186-/196	16-24
8	8	AS22759/83-/90, /186-/196	20-26
9	9	AS81044/6, /7	16-24
10	10	AS81044/6, /7	20-26
11	11	AS81044/12, /13, AS22759/40-/54	16-24
12	12	AS81044/12, /13	24-30
13	13	MIL-DTL-81381/1, /2, /6-/14, /17-/22	16-24

AS5768/ (*) – T1

Basic Spec Number ↑

Type of Tool (1- Custom Stripmaster) ↑
(2- Custom Stripmaster Lite) ↑

T1 = Tool & Blade Combination ↑
B1 = Blade Set Only ↑



Wire Stripping

Identification

- Tooling cross references for aerospace-grade wire can be found on the DMC website.
- These references include AS22759, AS81044, and other wire specifications.

CUSTOM STRIPMASTER® LITE WIRE-TO-TOOL CROSS REFERENCE			
This table is built using AS5768/2 Rev. D.			
Search Text			
Wire Range (AWG)	Wire Specification	Manual Tool w/ Blade	Replacement Blade Only
30-24	AS22759/9	45-640 AS5768/2-T2 (SAE)	LB-920 AS5768/2-B2 (SAE)
30-24	AS22759/10	45-640 AS5768/2-T2 (SAE)	LB-920 AS5768/2-B2 (SAE)
30-24	AS22759/11	45-640 AS5768/2-T2 (SAE)	LB-920 AS5768/2-B2 (SAE)
30-24	AS22759/12	45-640 AS5768/2-T2 (SAE)	LB-920 AS5768/2-B2 (SAE)
30-24	AS22759/13	45-640 AS5768/2-T2 (SAE)	LB-920 AS5768/2-B2 (SAE)
30-24	AS22759/14	45-640 AS5768/2-T2 (SAE)	LB-920 AS5768/2-B2 (SAE)
30-24	AS22759/15	45-640 AS5768/2-T2 (SAE)	LB-920 AS5768/2-B2 (SAE)
30-24	MIL-DTL-81381/3	45-640 AS5768/2-T2 (SAE)	LB-920 AS5768/2-B2 (SAE)

AS5768/1: dmctools.com/wire-to-tool-cross-reference

AS5768/2: dmctools.com/stripmaster-lite-wire-to-tool-cross-reference



Wire Stripping

General Information

Wire Stripping tools are supplied with Blades included; Blades sets can be ordered separately.

Using the 45-1987 as an example:

Style of Tool	Complete Tool with Blade	Spare Blade only	Wire Range AWG	Blade Type	AS5768 Tool with Blade	AS5768 Spare Blade
Custom Stripmaster	45-1987	45-1987-1 L-5561	16-26 26-30	Die Type	*AS5768/1-T9	*AS5789/1-B9
Custom Stripmaster Lite	45-2133 45-2118 45-640	45-2133-1 45-2118-1 LB-920	16-24 20-26 24-30	Die Type	**AS5768/2-T4 NA AS5768/2-T2	**AS5789/2-B4 NA AS5768/2-B2
Ergo-Elite Stripmaster	55-1987	55-1987-1	16-26	Die Type	NA	NA
Stripmaster	NA	NA	NA	Knife Type	NA	NA

Custom Stripmaster® blades use (-1) or (L-) Part Numbers

Custom Stripmaster Lite® blades use (-1) or (LB-) Part Numbers

Ergo Elite® blades use (-1) Part Numbers

*Custom Stripmaster ** Custom Stripmaster Lite





Wire Stripping

Identification

A common misconception is that two wires with the same gauge will have identical measurements and use the same tooling.

- Wire dimensions affect stripping requirements and tooling.
- Wire dimensions may differ by:
 - Gauge
 - Insulation type
 - Concentricity

Tooling & Application

Precision vs. Non-Precision

There are two basic levels of wire stripping, precision and non-precision.

- Precision

- Employed where extreme reliability and critical tolerances must be achieved.
- Required for aircraft, avionics, radar, and guidance systems
- Requires a die-type blade.

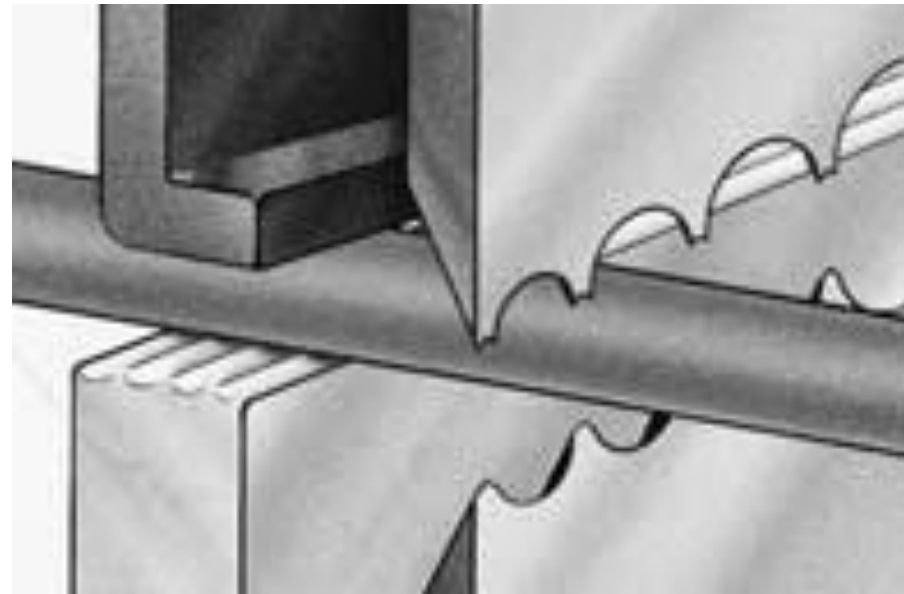
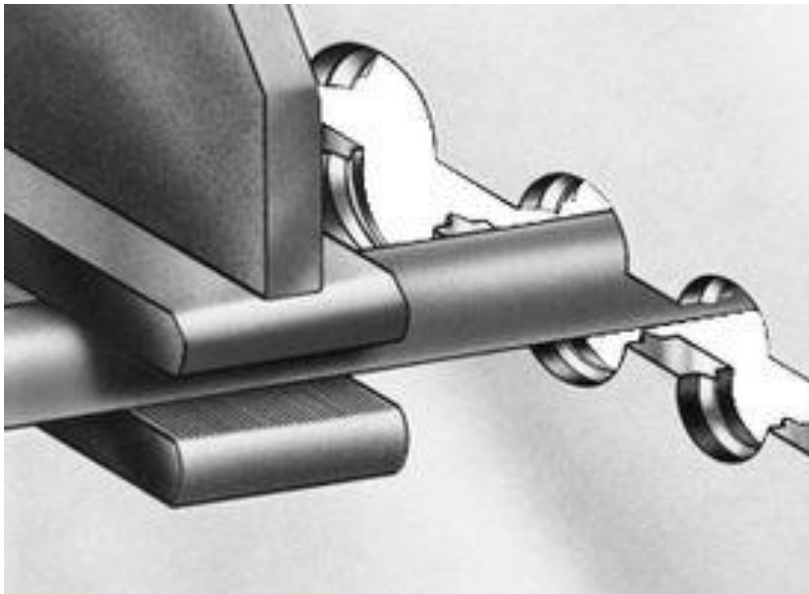
- Non-Precision

- Strip assures product reliability and can be obtained at a relatively low cost.
- Used in the manufacturing of lighting fixtures, electrical appliances, and automotive harnesses etc.
- Typically requires a knife-type blade.

Tooling & Application

Blade Information

- Die-type blades incorporate a counter-bored hole sized to the insulation and an inner cutting hole sized to the conductor. This varies depending on insulation type and thickness.



- Knife-type stripping blades have form-ground holes which penetrate thicker and oversized insulation like a knife.

Tooling & Application

Die-Type Blades

- **Custom Stripmaster®** precision wire strippers with **die-type** blades
 - Meets SAE-AS5768 standards
 - Mil-spec die-type blades allow precision stripping
 - Inner cutting hole sized to conductor and counter-bored holes sized to insulation
 - Strips 30 - 10 AWG solid or stranded wire
 - Cushioned, comfort grip handles
 - Replaceable blades



Red Handles (10-14 AWG)



Black Handles (16-30 AWG)



Tooling & Application

Die-Type Blades

- **Custom Stripmaster Lite[®]** precision wire strippers with **die-type** blades
 - 3/4 the size of a Custom Stripmaster[®]
 - Strips 16 - 30 AWG stranded wire
 - Requires 1/3 less hand pressure for each strip actuation
 - Blades are unique to this tool and are not interchangeable with Custom Stripmaster blades.
 - Meets SAE-AS5768 standards



Black Handles (16-30 AWG)

Tooling & Application

Die-Type Blades

- **Ergo-Elite®** Stripmaster wire stripper
 - The latest light-weight (5 oz.), ergonomically engineered wire stripper for the aerospace industry.
 - Tilted wire stripping head provides better leverage and clear view for wire positioning
 - Advanced jaw position ensures a quality strip on a wide range of wire gauges while eliminating potential scraping of the inner conductor and leaving a squared finish.
 - Advanced blade design has a built-in guiding syst for loading the wire properly into position.

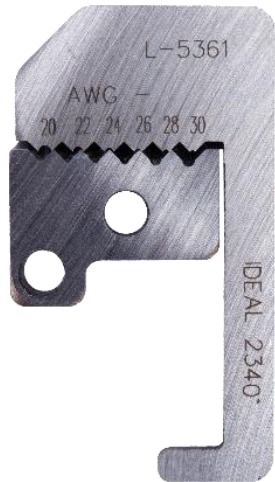


Tooling & Application

Knife-Type Blades

Stripmaster® non-precision wire strippers with knife-type blades

- Strips 30-6 AWG wire



T® Stripper non-precision tool

- Knife-type blades
- Varying AWG range



Tooling & Application

Accessories

- Adjustable Stop
 - Wire stops ensure uniform stripping lengths ranging from 3/32" to 3/4".
 - Stripmaster Lite: LB-1195
 - Custom Stripmaster: L-5270
 - Ergo Elite Stripmaster: LB-1904
- Short Stop Latches
 - Protects wire strands by limiting stripping stroke and retain the slug until the wire is ready for terminating.
 - Custom Stripmaster: L-5269
 - Stripmaster Lite: LB-1194

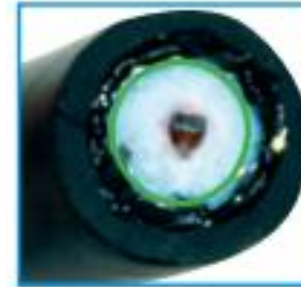


Tooling & Application

Knife-Type Blades

T[®]-Cutter Lite are non-precision tools with knife-type blades.

- They feature a lightweight, compact design.
- Cushioned grips.



Tooling & Application

Non-Precision Wire Strippers



Tooling & Application

Insulation Removal



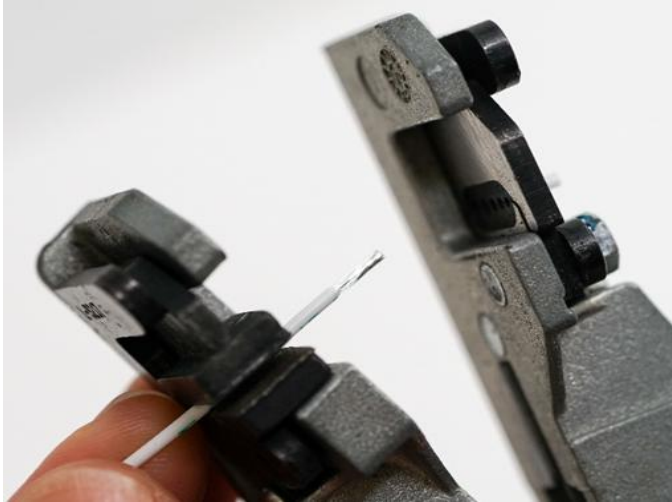
Insert the wire to be stripped into the correct slot for the wire size (each slot is marked with wire size). Make sure you insert the wire into the center of the slot.



Close the handles slowly to seat the wire properly into the correct blade slot and close the grip against the wire.

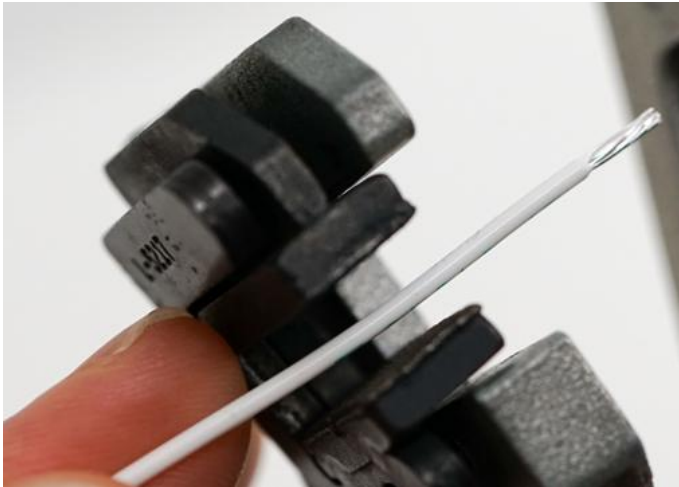
Tooling & Application

Insulation Removal



Close the handles together as close as possible to remove the insulation.

Partially open the handles to allow the grips to open.



To prevent “bird caging” of wire strands, remove the stripped wire before letting the handles open completely.

Release the handles to allow the wire holder to open.

Wire Stripping

Inspection

Incorrect wire stripping practices can lead to wire failure, short circuits, or damage to equipment. Damage to wires should not exceed the limits specified in the table.

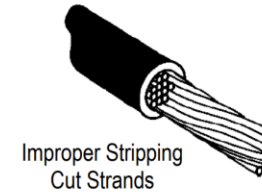
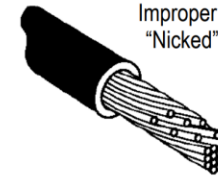


ACCEPT



REJECT

Improper Stripping
"Nicked" Strands



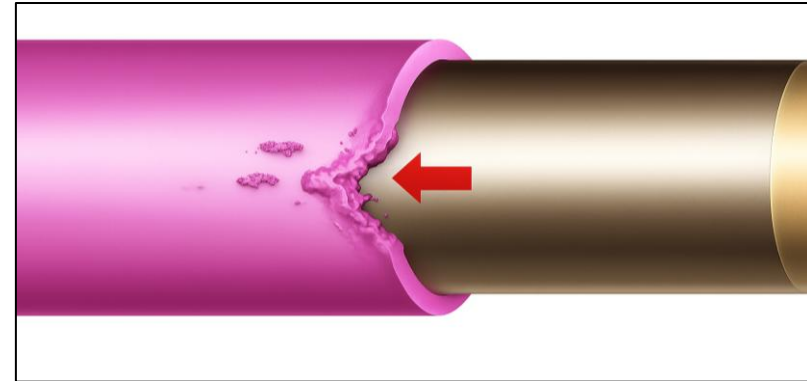
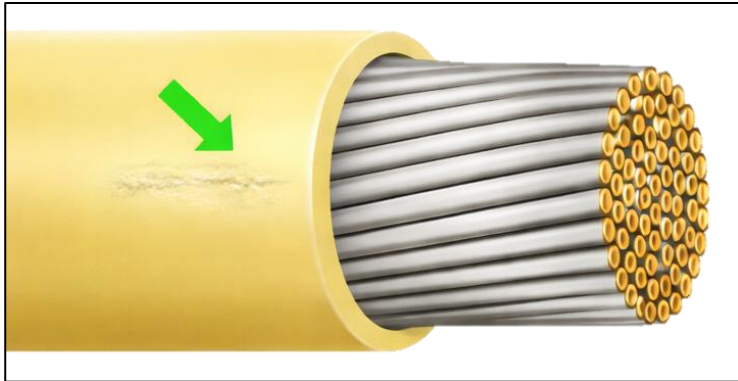
Maximum allowable nicked and broken strands			
Wire Size	Conductor material	Number of strands per conductor	Total allowable nicked and broken strands
24-14	Copper or Copper Alloy	19	2 nicked, none broken
12-10		37	4 nicked, none broken
8-4		133	6 nicked, 6 broken
2-1		665-817	6 nicked, 6 broken
0-00		1,045-1,330	6 nicked, 6 broken
000		1,665-	6 nicked, 6 broken
0000		2,109-	6 nicked, 6 broken
8-000	Aluminum	All numbers of strands	None, None

Source: AC43.13

Wire Stripping

Inspection

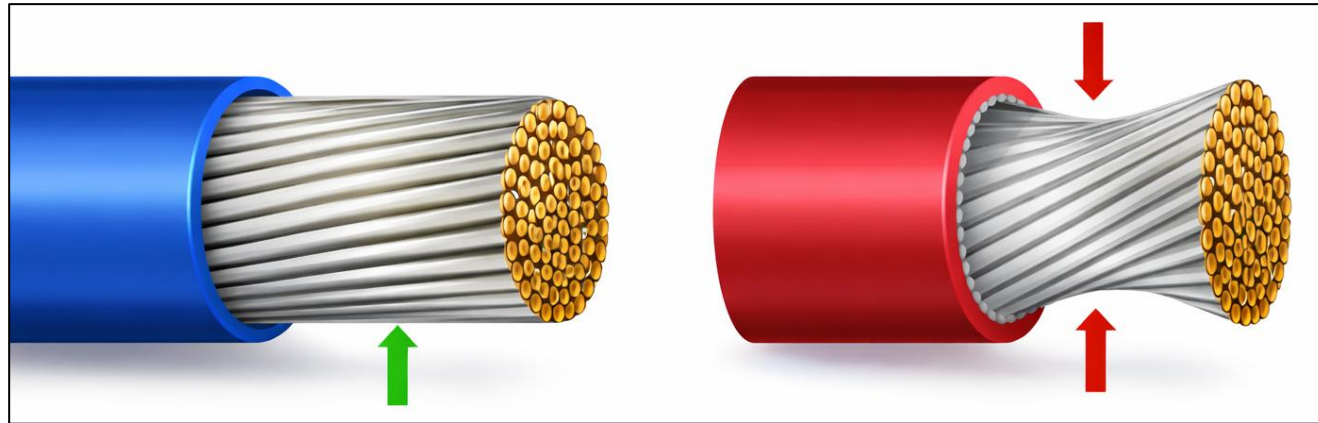
- If you use the incorrect blade for the wire insulation type, it may result in damage.
- Insulation shouldn't have nicks, cuts, or other damage.
- Slight scuffing (dull/rubbed appearance) of the insulation surface finish is acceptable.
- Don't use insulation that's damaged beyond minor scuffing.



Wire Stripping

Inspection

- If the twist pattern (lay) of the wire strands is disturbed, restore it to the original pattern as closely as possible. Re-twisted lay is acceptable when no other damage is evident.
- Strands twisted beyond the normal twist pattern put more stress on the individual strands and may result in conductor breakage.



Tooling & Application

Calibration Procedures for Custom Stripmaster and Custom Stripmaster Lite wire strippers



Step 1

Place the lower stationary blade half over the blade mounting studs.



Step 2

The blade dogleg should be attached first then depressed to seat itself. Slightly depress blade handle to allow the blade to be properly seated. Insert the dogleg of the upper movable blade into the locating slot of the side latch.



Step 3

Position the blade retaining plate over the mounting studs. Install split lock-washers and hex nuts onto the mounting studs. Do not tighten hex nuts at this time.

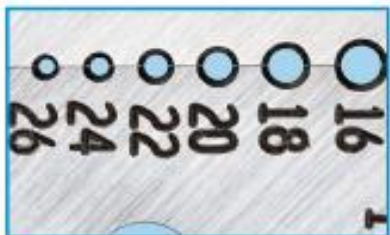


Step 4

First, position the lower blade half cutting surface in line with the lower gripper pad surface by slightly depressing the gripper handle.

Next, depress both handles just enough to allow the top and bottom blade surfaces to meet. (Exerting too much pressure on the handles could

force the bottom blade out of alignment.) Now tighten the hex nuts to a maximum torque value of 10 in. lbs.



Step 5

Visually check the upper and lower blade alignment by depressing the blade handle only until the blade surfaces make contact. Leaving the gripper

pads in the open position, look at the blade cutting holes and mating surface between holes. This is most effectively accomplished with a light source behind the stripper. Look for light in cracks between the cutting holes. If light can be seen in these areas, inspect the blades' mating surfaces for debris or imperfections. Clean the blades and check for light "leaks" again. If the blades have debris or imperfections, set-up was not correctly performed. In this case, repeat the set-up procedures.

Step 6

The final checking procedure to confirm proper blade set-up and calibration is to perform a series of strips on wire samples for which the blades were designed. In doing so, be sure the one-step stripping action is smooth with positive slug removal. During the stripping process check that the wire does not slip in the grippers and that the wire is not excessively bent by the gripper pads. Finally, closely inspect the stripped wire to be certain there are no nicks, scrapes or other defects in the conductor.

Should any of the above occur, repeat the blade set-up and calibration procedures. Also, be certain that the sample wire is indeed correctly matched with the cutting blades used. Refer to the blade reference charts on pages 20 through 22 for blade selection guidelines.

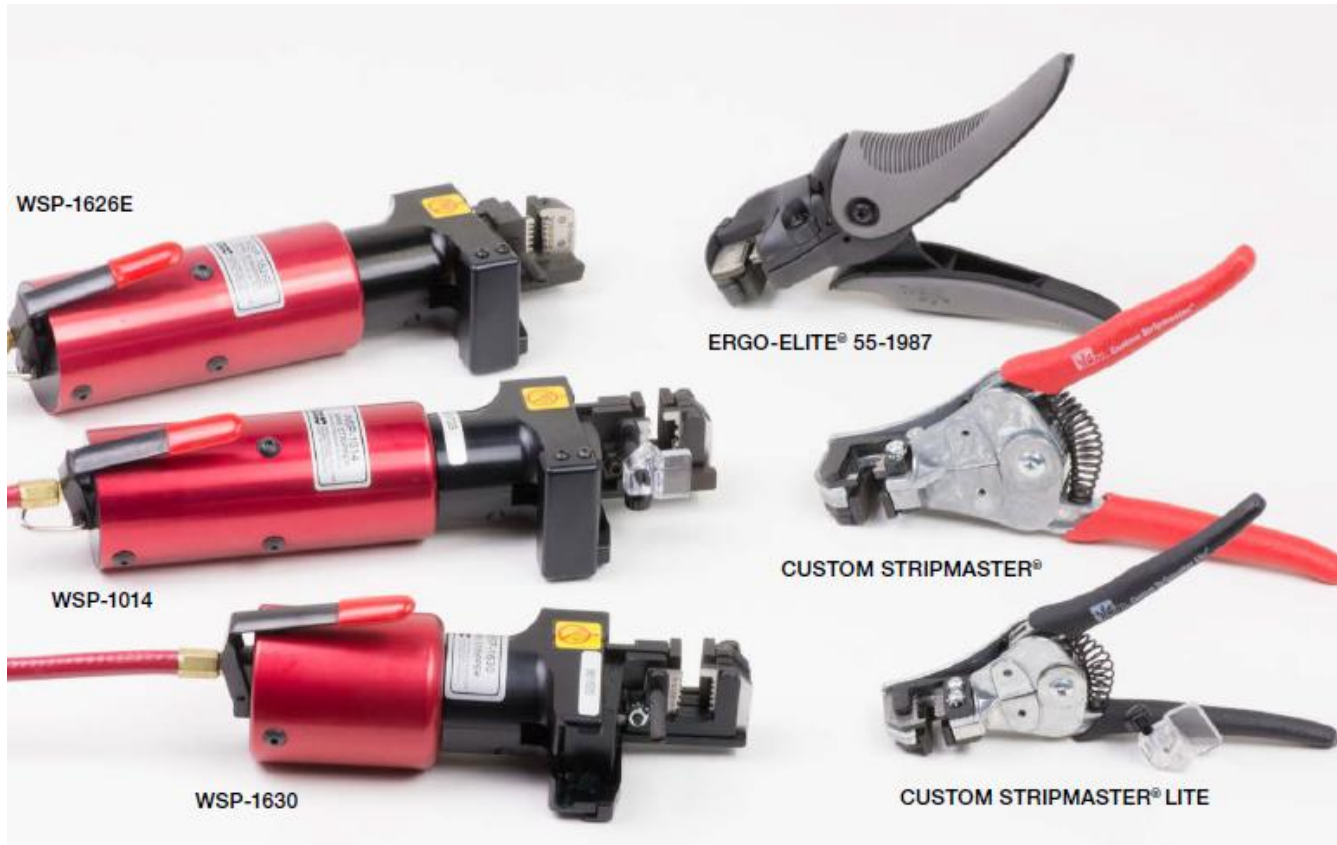
Caution: Never plug or mechanically gauge blade holes as this could result in damage to the blade.

With the cutting blades properly set-up and calibrated, the Custom Stripmaster® and Standard Stripmaster® Wire Strippers provide automatic, repeatable and superior quality wire stripping in a wide variety of production applications. Quality control and efficiency are assured.

Tooling & Application

Pneumatic Tools

Pneumatic tools utilize blades and accessories that are approved for AS5768.

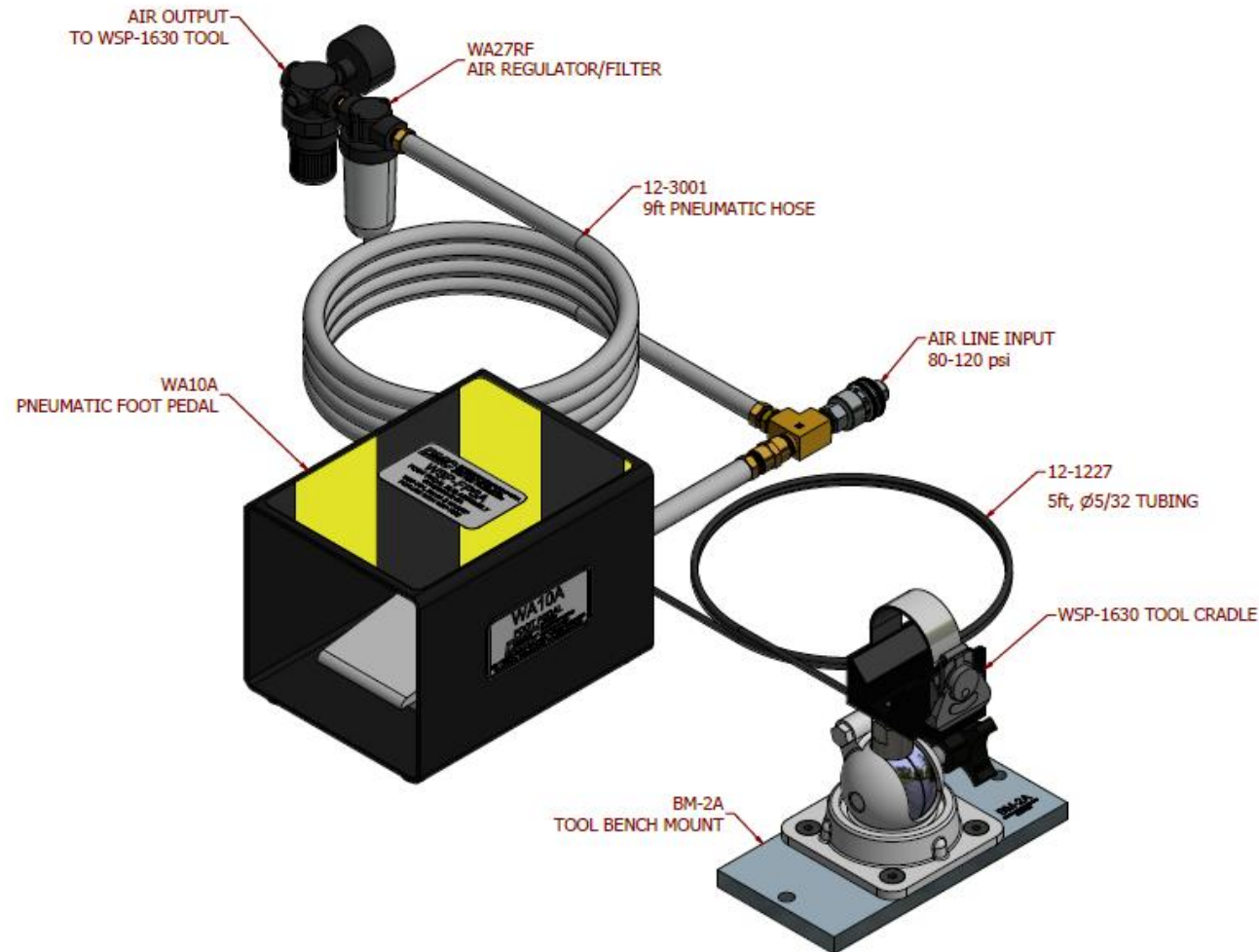


- DMC offers a line of pneumatic wire strippers that are equivalent to the mechanical versions, Custom Stripmaster and Ergo Elite.
- 360-degree rotatable stripping head
- Flow control valve to adjust stripping speed
- Operates on standard air pressure of 80-120 psi (5.4-8.2 bar).

Tooling & Application

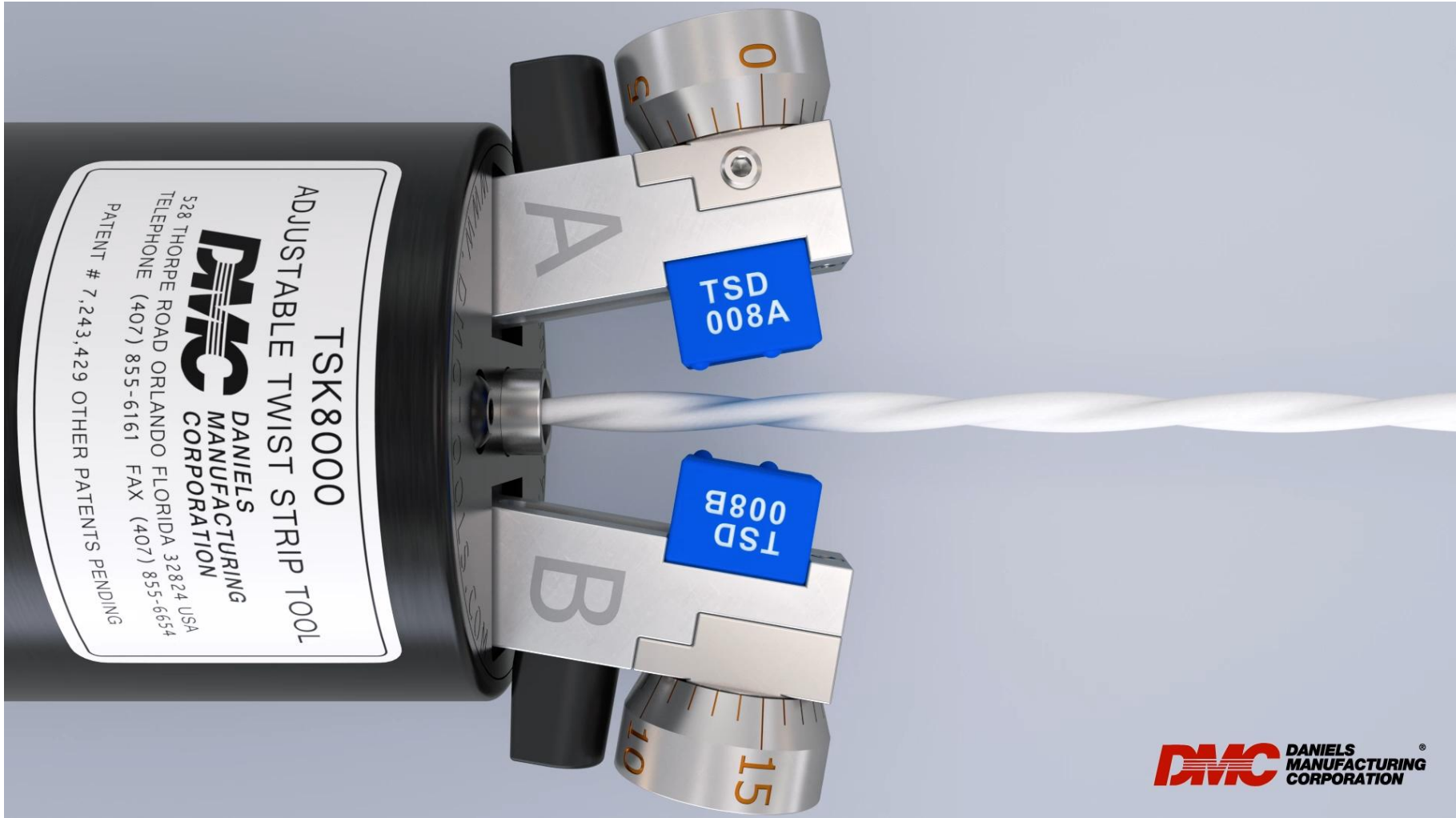
Pneumatic Tools

DMC offers a compatible bench mount and foot pedal (WSP-FPSA) for our pneumatic wire strippers.



Tooling & Application

Twist-Strip®



Tooling & Application

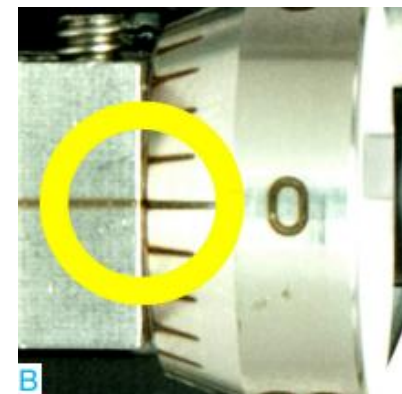
Twist-Strip®

- Twist-Strip® outer jacket wire stripper is meticulously engineered to simplify the challenges posed by traditional methods of stripping multi-conductor, jacketed shielded, twisted, and round cables.
- Twist-Strip tools accommodate single, double, triple and quad wire configurations ranging from 0.081" - 0.310".
- The TSK8000 (0.081- 0.180) is qualified to MIL-DTL-32610.
- The TSK8500 (0.181- 0.310) covers the larger range of wires.



Tooling & Application

Twist-Strip®



A cable track is precisely machined into each die set. This contoured track creates a controlled rotation of the dies/blades as the cable is moved along the track.

Twist-Strip has adjustable blade depth settings.

The die container recommends a starting depth setting for die half "A & B" (ex: B=.010")

Turn depth control knob on the tool clockwise, each hash mark on the knob represents .001" of blade travel.



Tooling & Application

Twist-Strip®

- Dies are available for Single, Double, Triple and Quad wire configurations.
 - First, begin by determining the cable lay. If the cable twists in an “S” direction it is a left lay. If the cable twists in a “Z” direction it is a right lay.
 - Next, use the cable diameter gauge card included with your tool to find the smallest opening that the wire can pass through freely. The gauge card will indicate the corresponding die set that should be used.



DIE SET CONTAINER



TSK8000 (KIT)
DIE SETS SOLD SEPARATELY



Tooling & Application

Twist-Strip®

1

IDENTIFY

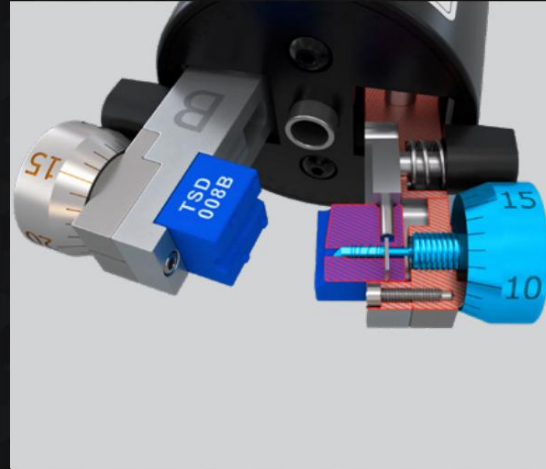
Identify the correct die set with the cable diameter gauge card.



2

ADJUST

Adjust the depth setting of the concealed carbide blades.



3

INSERT

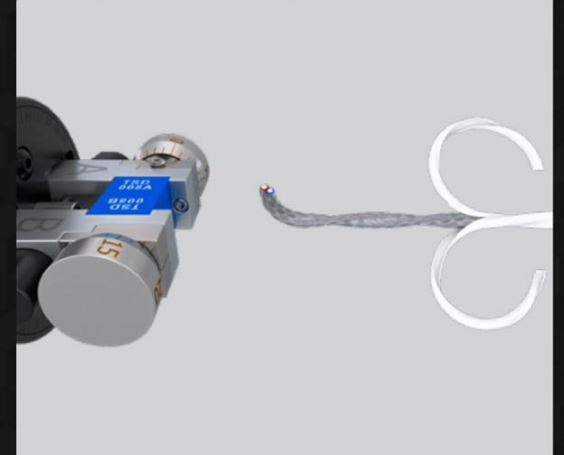
Squeeze the arm levers to insert the cable and release to close the arms.



4

STRIP

Pull the tool away from the cable to produce a light score. Slightly bend the cable and remove the jacket halves.



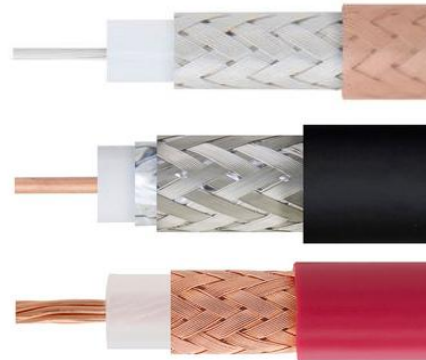
Wire Stripping

Coaxial Wire Strippers

- Coaxial adjustable blade strippers are available in four models covering cable diameters from 1/8" up to 9/16".
- All models include two adjustable depth blades on opposing sides, one adjustable slitting blade on the nose.
- Coaxial Cables can be stripped using Stripmaster tools and two step blades designed to remove the outer jacket and dielectric.



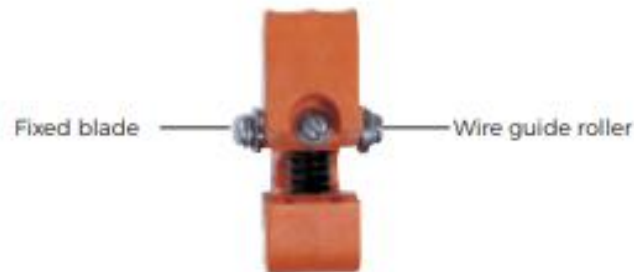
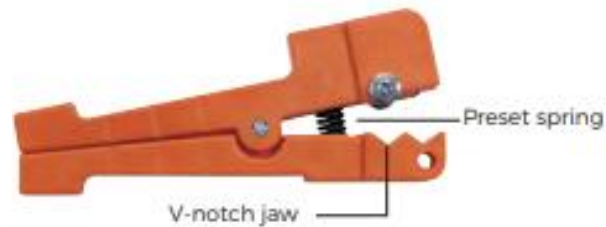
RG8, RG11, RG-174, RG-187,
RG-58, RG-59/6, UTP Cable



Wire Stripping

Ringer Shielded Cable Strippers

- Precision stripping for most non-round shielded cable and other outer cable jackets, including extruded PTFE insulations.
- Spring loaded cutting head holds cable with consistently accurate tension, removing any need for adjustment.
- V-notch jaw automatically positions and holds cable, increasing cutting accuracy.
- Fixed blades are sized to insulation thickness, preventing inaccurate cuts, and 17-4 stainless steel construction assures long life.
- Wire guide roller maintains stripping head alignment, producing a square, clean cut.



Wire Stripping

Ringer Shielded Cable Strippers

- Ringers come in five colors for easy identification.
- Each tool body has a unique V notch to accommodate wire ranging from .012 - .030.



45-400
For cable
diameters up to
3mm (0.12"). Blades
sold
separately - see list
on next page



45-401
For cable
diameters from
3mm (0.12") to 5.5
mm (0.22") with
5 Mil insulation
thickness,
including Kapton®.
Comes with K-6492
blade



45-402
For cable
diameters from
3mm (0.12") to
5.5 mm (0.22") with
8-10 Mil insulation
thickness,
including Kynar®.
Comes with
K-6493 blade



45-403
For cable
diameters from
3mm (0.12") to
5.5 mm (0.22").
Blades sold
separately - see list on
next page



45-404
With "deep-V" for
larger cable
diameters up to
7.6mm (0.30")
Blades sold
separately - see list
on next page

Wire Stripping

Ringer Blades

- For use with any ringer shielded cable stripper.
- Select a blade with a cutting depth less than the insulation thickness.

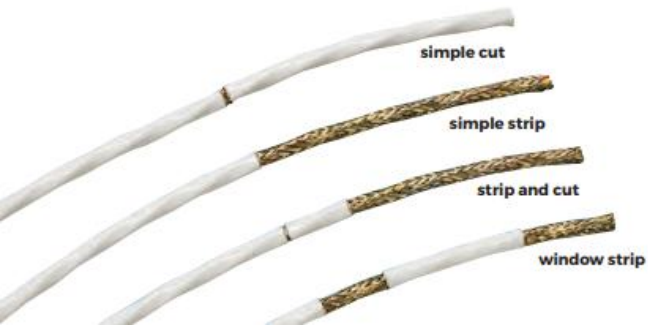


Cutting Depth mm/in.	Cat. No.
0.076mm / .0030 in.	K-6491
0.112mm / .0044 in.	K-6492
0.152mm / .0060 in.	K-6493
0.178mm / .0070 in.	K-6494

Cutting Depth mm/in.	Cat. No.
0.203mm / .0080 in.	K-6495
0.229mm / .0090 in.	K-6496
0.254mm / .0100 in.	K-6497
0.279mm / .0110 in.	K-6498
0.305mm / .0120 in.	K-6499
0.330mm / .0130 in.	K-6500

Cutting Depth mm/in.	Cat. No.
0.356mm / .0140 in.	K-6501
0.406mm / .0160 in.	K-6502
0.457mm / .0180 in.	K-6503
0.508mm / .0200 in.	K-6504
0.559mm / .0220 in.	K-6505
0.635mm / .0250 in.	45-2108-1

- Adjustable wire stop
 - Provides uniform stripping length.



LB-825, 3/8 in. to 2-1/2 in.
LB-840, 2-1/2 in. to 5 in.

Wire Stripping

Conclusion

- High reliability wire stripping is engineered, not improvised. It starts with selecting the right tooling, following proven procedures and knowing exactly what defects to inspect for before failures ever reach the field.
- No matter which type of wire stripper you use, the key features to look for in avionics-grade wire strippers are:
 - Precision machined cutting edges that stay sharp
 - Adjustable stop for consistent strip lengths
 - Comfortable ergonomic handles for extended use
 - Rugged construction to withstand frequent use
- DMC brings 77 years of tooling expertise, and we'd welcome the opportunity to support your next project.





**Thank
You**

